

No m -cycles of the $3n-1$ map for $m \leq 49$

The Simons–de Weger template on the negative side, from a verification floor of 2^{44}

Philippe Cochin
philippe@cochin.fr

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Laboratory manuscript, first draft of 21 September 2026. Not a review object. One external input, Rhin’s effective measure for linear forms in $\log 2$ and $\log 3$; one computational input, the laboratory’s own verification floor; everything else is proved below or is exact arithmetic recorded with its margins. Simons–de Weger 2005 and Rhin 1987 were read in full on the day of this draft, and the constants below are compared with theirs where the comparison is informative.

Abstract

Let $g(y) = y/2$ for even y and $g(y) = (3y-1)/2$ for odd y , the $3n-1$ map on the positive integers, equivalently the shortcut $3n+1$ map on the negative integers. Its known cycles are 1, (5, 7, 10) and the eleven-element cycle at 17, and every start below 2^{44} reaches one of them. An m -cycle is a cycle with m maximal runs of odd elements. We transpose the template of Simons and de Weger for $3n+1$ m -cycles to this map, deriving its constants on this side rather than borrowing them: the odd step subtracts, so in the variable $u = y-1$ an odd run is exact multiplication by $3/2$, a run of a odd steps starts at $y \geq 2^a + 1$, the cycle equation bounds the linear form $\Lambda = o \log 3 - K \log 2$ by $m/(x_{\min} - 1)$ with constant one, and successive local minima obey $u_{i+1} < u_i^{\log_2 3}/2$. With Rhin’s bound this gives: the $3n-1$ map has no m -cycle with $1 \leq m \leq 49$ other than the two known ones. For $m \leq 43$ no admissible cycle length lies below Rhin’s ceiling; for $44 \leq m \leq 49$ the admissible lengths are excluded by the chaining, the closest by 1.4 bits. At $m = 50$ four lengths remain, the input for lattice reduction. The same tables give $m \leq 44$ from 2^{40} , $m \leq 68$ from 2^{60} and $m \leq 82$ from 2^{68} . No published verification floor and no m -cycle theorem with $m \geq 3$ existed for this map.

1. The map, the cycles, and what is known

Write g for the shortcut $3n-1$ map above. Its orbits on the positive integers are the orbits of the shortcut Collatz map $T(x) = x/2, (3x+1)/2$ on the negative integers under $x = -y$: $(3(-y)+1)/2 = -(3y-1)/2$. Three cycles are known,

$$(1), \quad (5, 7, 10), \quad (17, 25, 37, 55, 82, 41, 61, 91, 136, 68, 34),$$

with $(K, o) = (1, 1), (3, 2), (11, 7)$ steps and odd steps. Whether these are all the cycles, and whether every orbit is bounded, are the $3x-1$ questions of Lagarias’s bibliographies, open.

The floor. Every $1 \leq y < 2^{44}$ reaches one of the three cycles. This is the laboratory’s own computation (19–20 September 2026), descent by strong induction on odd starts in one hundred and twelve chunks: sixteen with a plain walker through 2^{40} (greatest step count 544 against a cap of 4000), then ninety-six chunks of $5 \cdot 2^{35}$ starts with a residue sieve modulo 2^{24} and a 2^{16} -step block map as an accelerator, 8246337208320 odd starts counted exactly (greatest step

count 704 against a cap of 40000, counted in blocks of sixteen); the seven odd starts the first sixteen chunks had skipped, found from their own printed counts, are verified by full iteration. No start failed and no new cycle appeared. The verifier sources, the chunk reports and the coverage check are archived with the branch `negative_floor_3x1`. No published floor exists: the only statement in print is an unattributed comment on OEIS A037084, “up to at least 100000000”. The certificate is a computation and is labelled as one throughout.

m -cycles. A cycle with at least one even step splits into $m \geq 1$ maximal runs of odd elements alternating with runs of even elements. The first element of an odd run is a local minimum, the element after its last odd step a local maximum. The two known cycles with an even step have $m = 1$ and $m = 2$. Simons [S07] proves that g has exactly one nontrivial 2-cycle, at 17, and does so without any floor, by de Weger’s bounds on $0 < 3^K - 2^{K+L} < 3^{0.89K}$, Steiner’s 1-cycle theorem and an extremal inequality; he states that the floor-free route stops at $m \geq 3$. For $3n + 1$, Simons [Si05] excludes 2-cycles, Simons and de Weger [SdW] exclude m -cycles for $m \leq 68$ from the floor of November 2004, $X_0 = 301 \cdot 2^{50} > 3.3889 \cdot 10^{17}$, and Hercher [H23] reaches $m \leq 91$ from $695 \cdot 2^{60}$ with the arrangement of the valleys. The template of [SdW] has five steps: a floor; a bound on Λ from the cycle equation (their Lemma 4 and Corollary 5); a chaining of the local minima (Lemmas 6, 7); Crandall’s lemma generalized for a lower bound on the length (Lemma 10, Corollary 11) against Rhin’s bound for an upper one (Lemmas 12, 14), sharpened through the partial quotients of $\log_2 3$ (Lemma 16); and de Weger’s approximation lattice where a window remains (Lemma 18). At their floor the three stages give $m \leq 57$, $m \leq 63$ and $m \leq 68$. Nobody has run any of it on $3n - 1$, because the first step was missing; [S08], which extends the template to $3x + q$, $px + q$ and Guy’s permutation, skips the map. This note runs the first four steps with the constants of this side, replaces the partial-quotient stage by an exact enumeration of the admissible lengths, and leaves the lattice as its open list.

2. Notation

Throughout $x = \log 2 / \log 3$, $\delta = \log_2 3 = 1/x$. For a cycle C of g with K steps and o odd steps put

$$\Lambda(C) = o \log 3 - K \log 2.$$

For odd y put $u = y - 1$. The local minima of an m -cycle are $y_1, \dots, y_m$ in cyclic order, $u_i = y_i - 1$, the run from y_i has a_i odd steps, and $r_i \geq 1$ halvings follow it. $x_{\min}$ is the least element of C . The floor is $X_0 = 2^{44}$.

3. Four lemmas on the negative side

Lemma 1 (odd runs). Let $y \geq 3$ be odd, $u = y - 1$, $a = v_2(u)$. Then

$$g^k(y) - 1 = \left(\frac{3}{2}\right)^k (y - 1) \quad (0 \leq k \leq a),$$

$g^k(y)$ is odd for $k < a$ and $g^a(y)$ is even. Consequently a run of a odd steps starts at $y \geq 2^a + 1$, and its local maximum $M = g^a(y)$ satisfies $M \leq u^\delta + 1$.

Proof. For odd y , $g(y) - 1 = (3y - 1)/2 - 1 = \frac{3}{2}(y - 1)$. While the iterates stay odd this repeats, so $g^k(y) = 1 + 3^k u / 2^k$ as long as $g^j(y)$ is odd for $j < k$; and $1 + 3^k u / 2^k$ is an odd integer exactly when $2^{k+1} \mid u$, since 3^k is odd. Hence the first $a = v_2(u)$ iterates are odd and the next is even. Then $u \geq 2^a$ gives $y \geq 2^a + 1$, and $(3/2)^a \leq (3/2)^{\log_2 u} = u^{\log_2(3/2)} = u^{\delta-1}$ gives $M = 1 + (3/2)^a u \leq 1 + u^\delta$. $\square$

This is Hercher’s Lemma 8 [H23] with $y \equiv 1 \pmod{2^a}$ in place of $y \equiv -1$: the floor $2^a + 1$ is larger by two, and it is attained at 5 ($a = 2$) and 17 ($a = 4$).

Lemma 2 (cycle equation). On a cycle C with at least one even step,

$$\Lambda(C) = - \sum_{y \in C \text{ odd}} \log\left(1 - \frac{1}{3y}\right) > 0, \quad \text{so} \quad 3^o > 2^K,$$

and $\Lambda(C) < \sum_{y \in C \text{ odd}} \frac{1}{3y-1}$. The run from a local minimum y contributes less than $1/(y-1)$. Hence, for an m -cycle,

$$\Lambda(C) < \sum_{i=1}^m \frac{1}{y_i - 1} \leq \frac{m}{x_{\min} - 1}.$$

Proof. An odd step multiplies by $\frac{3}{2}(1 - \frac{1}{3y})$ and an even step by $\frac{1}{2}$; around the cycle the product is 1, so $3^{o/2-K} \prod_{\text{odd}} (1 - 1/(3y)) = 1$, which is the displayed identity, and the product is below 1. The bound $-\log(1-t) \leq t/(1-t)$ at $t = 1/(3y)$ gives $1/(3y-1)$. Along the run from y , Lemma 1 gives $3g^k(y) - 1 = 3(3/2)^k u + 2 > 3(3/2)^k u$, so the run contributes less than $\frac{1}{3u} \sum_{k \geq 0} (2/3)^k = 1/u$. Finally the least element of C is odd (an even least element would halve to something smaller) and its predecessor is even (an odd predecessor y' would have $y' < (3y' - 1)/2 = x_{\min}$), so $x_{\min}$ is a local minimum and every $y_i \geq x_{\min}$. $\square$

This is [SdW] Lemma 4 and Corollary 5 with the sign flipped. The corresponding bound on the positive side, Hercher's Corollary 17, carries the constant $97/54$, and [SdW] Lemma 4 has $\sum 1/x_i$ over the minima themselves. Here it is $1/(y_i - 1)$: the odd step subtracts, so the run's elements sit at or above the geometric progression rather than below it.

Lemma 3 (chaining). For consecutive local minima, $u_{i+1} < u_i^\delta/2$. Hence, with $B(m) = (\delta^m - 1)/(\delta - 1)$,

$$o = \sum_{i=1}^m a_i \leq B(m) \log_2 u_1 - \frac{B(m) - m}{\delta - 1}, \quad K < \delta o,$$

and therefore, taking $y_1 = x_{\min}$,

$$\log_2(x_{\min} - 1) > L_{\min}(K, m) := \frac{1}{B(m)} \left(\frac{K}{\delta} + \frac{B(m) - m}{\delta - 1} \right).$$

Proof. $y_{i+1} = M_i/2^{r_i}$ with $r_i \geq 1$ and $M_i \leq u_i^\delta + 1$ (Lemma 1), so $u_{i+1} = M_i/2^{r_i} - 1 \leq (u_i^\delta + 1)/2 - 1 < u_i^\delta/2$. Taking logarithms, $\log_2 u_{i+1} < \delta \log_2 u_i - 1$, and by induction $\log_2 u_i \leq \delta^{i-1} \log_2 u_1 - (\delta^{i-1} - 1)/(\delta - 1)$. Summing over i and using $a_i = v_2(u_i) \leq \log_2 u_i$ gives the bound on o ; $K \log 2 < o \log 3$ is $\Lambda > 0$. Solving for $\log_2 u_1$ gives $L_{\min}$. $\square$

The chaining is tight on the 17-cycle: $u_1 = 16$, $u_2 = 40$, and $16^\delta/2 = 40.5$. It is [SdW] Lemma 6 in the variable u : their $x_{i+1} < b^\delta x_i^\delta$ with $b = (1 + X_0^{-1})/2^{1/\delta}$ becomes exact here, with no X_0 in the constant, because $u = y - 1$ is the conjugated variable itself. The bound on o is their Lemma 7 read as a lower bound on the least element rather than as an upper bound on Λ .

Lemma 4 (admissible lengths). If an m -cycle has $x_{\min} \geq X_0$, then $0 < \Lambda < m/(X_0 - 1) < \log 3$, so $o = \lceil Kx \rceil$ and

$$\Lambda(K) = \log 3 (1 - \{Kx\}) < \frac{m}{X_0 - 1}.$$

The lengths K satisfying this are exactly those with $1 - \{Kx\} < m/((X_0 - 1) \log 3)$, and below any bound they are enumerated exactly by the three-gap walk: from the two least returns q_1, q_2 of $\{Kx\}$ into the two windows, found among the convergents and semiconvergents of x , every further member is the previous one plus q_1, q_2 or $q_1 + q_2$. The least member is Crandall's lower bound on the length.

Proof. Lemma 2 and $x_{\min} \geq X_0$. The walk is the classical three-distance structure of $\{Kx\}$ and is the laboratory's `three_gap_walk`, checked against a direct scan at coarse thresholds. $\square$

[SdW] reach the same set from above and below: Lemma 10 and Corollary 11 give the least convergent denominator the floor admits, and Lemma 16 caps the length through the largest partial quotient below that index. The exact enumeration replaces both, and, at their floor, lands exactly on their Lemma 17 (Section 5).

4. Rhin's ceiling

Proposition 5. Rhin's Proposition [R87, p. 160, (7)] states that for integers u_0, u_1, u_2 with $H = \max(|u_1|, |u_2|) \geq 2$,

$$|u_0 + u_1 \log 2 + u_2 \log 3| \geq H^{-13.3},$$

with no further constant. For an m -cycle of length $K \geq 2$ with an even step, $(u_0, u_1, u_2) = (0, -K, o)$ and $H = K$, so

$$2^{L_{\min}(K, m)} < x_{\min} - 1 < \frac{m}{\Lambda} \leq m K^{13.3}.$$

Let $K_3(m)$ be the least integer such that $2^{L_{\min}(K, m)} \geq 2m K^{13.3}$ for every $K \geq K_3(m)$. Then every m -cycle has $K < K_3(m)$.

Proof. Lemmas 2 and 3 with Rhin's bound. The function $K \mapsto L_{\min}(K, m) - 13.3 \log_2 K - \log_2(2m)$ is convex, negative at $K = 2$, and has a larger root; beyond it the function increases, so $K_3(m)$ is well defined, and the doubled constant covers $m/\Lambda + 1$. $\square$

Rhin's bound is the one external input of this note. It is a theorem, not a hypothesis; it is external in the sense that nothing here re-proves it and no Lean statement carries it. [SdW] Lemma 12 states it as $\Lambda > e^{-13.3(0.46057 + \log K)}$ in their odd count K , which is (7) at $H = K + L < 1.000001 \delta K$ with $0.46057 = \log \delta$; the laboratory's Paper A and its cycle-gap branch carried that form, weaker by the factor $\delta^{13.3} = 457$ when H is the length itself, and it is (7) that is used here. The general two-logarithm bound of Laurent–Mignotte–Nesterenko [LMN] would replace 13.3 by an exponent in the thousands at these heights and is not competitive; Rhin's constant is specific to $\log 2$ and $\log 3$.

5. The theorem

Theorem 6. Let $X_0 = 2^{44}$. For every $1 \leq m \leq 49$ and every length $K < K_3(m)$ with $\Lambda(K) < m/(X_0 - 1)$,

$$2^{L_{\min}(K, m)} \geq \frac{m}{\Lambda(K)}.$$

Consequently, with Rhin's bound, the $3n - 1$ map has no m -cycle with $1 \leq m \leq 49$ whose least element is at least 2^{44} , and by the floor no m -cycle with $m \leq 49$ other than $(5, 7, 10)$ and the cycle at 17.

Proof. An m -cycle above the floor has $K < K_3(m)$ by Proposition 5, its length is admissible by Lemma 4, and Lemmas 2 and 3 give $2^{L_{\min}(K, m)} < x_{\min} - 1 \leq m/\Lambda(K)$, contradicting the display. The display is a finite exact computation: for each m the admissible lengths below $K_3(m)$ are enumerated by Lemma 4 and each is tested at eighty digits. For $m \leq 43$ there is no admissible length below $K_3(m)$ at all, so the conclusion there rests on Lemmas 2 and 4 and Proposition 5 alone. For $44 \leq m \leq 49$ the admissible lengths exist and fail the display by the margins of Table 1. $\square$

Table 1. Rows at the floor 2^{44} . “Margin” is $\max \log_2(m/(\Lambda(K) 2^{L_{\min}}))$ over the admissible $K < K_3(m)$, negative when every admissible length is excluded.

m	$m/(X_0 - 1)$	$K_3(m)$	admissible $K < K_3$	least admissible	margin (bits)
1	$5.7 \cdot 10^{-14}$	155	0	—	—
2	$1.1 \cdot 10^{-13}$	495	0	—	—
5	$2.8 \cdot 10^{-13}$	3926	0	—	—
10	$5.7 \cdot 10^{-13}$	57129	0	—	—
20	$1.1 \cdot 10^{-12}$	8393107	0	—	—
30	$1.7 \cdot 10^{-12}$	1094917645	0	—	—
40	$2.3 \cdot 10^{-12}$	134716197551	0	—	—
44	$2.5 \cdot 10^{-12}$	913288455197	1	757698850864	−399.7
45	$2.6 \cdot 10^{-12}$	1472498495287	3	757698850864	−236.4
47	$2.7 \cdot 10^{-12}$	3824399289665	9	757698850864	−68.3
48	$2.7 \cdot 10^{-12}$	6160750059301	15	757698850864	−27.3
49	$2.8 \cdot 10^{-12}$	9921719447060	25	757698850864	−1.4
50	$2.8 \cdot 10^{-12}$	15974496522785	41	539722056247	+22.4

The exclusion at $m = 49$ hangs on 1.4 bits at $K = 757698850864$; any weakening of the constants, in Rhin’s form or in the chaining, would move the theorem to $m \leq 48$. At $m = 50$ the template leaves four lengths,

$$539722056247, \quad 757698850864, \quad 975675645481, \quad 1193652440098,$$

with 22.4, 15.0, 7.8 and 2.9 bits of room; they are the near-convergent lengths of x on the expanding side, and they are the input for the fifth step of the template, lattice reduction on the m local minima, which this note does not perform.

Table 2. What each floor buys, by the same tables.

floor	m excluded through	first open m	lengths left at the first open m
2^{40}	44	45	72448885240, 82888745831, 93328606422, 103768467013
2^{44}	49	50	539722056247, 757698850864, 975675645481, 1193652440098
2^{48}	49	50	1193652440098
$301 \cdot 2^{50}$ ([SdW]’s floor)	63	64	766512153894657
2^{60}	68	69	9881527843552324
2^{68}	82	83	6094436882695943503, 6724555128221608268

Two comparisons calibrate the method, on the other sign and so as plausibility checks rather than reproductions. At [SdW]’s own floor, $301 \cdot 2^{50}$, their three stages give $m \leq 57$ by the tables, $m \leq 63$ after the continued-fraction reduction and $m \leq 68$ after the lattice; the exact enumeration here gives 63 at that floor, which is their Lemma 17 exactly, so the lattice stage is worth five values of m there and would be worth a comparable amount at 2^{44} . Hercher’s 91 at $695 \cdot 2^{60}$ against the plain template’s 82 at 2^{68} prices his valley arrangement at about nine values of m , which is therefore the second refinement to transpose. The floor enters only through the admissibility threshold, in steps: 2^{48} buys nothing over 2^{44} except one survivor fewer at $m = 50$.

6. The cycles that exist pass the same test

The inequalities were checked where they must not exclude. With the floor set at 17 and $m = 2$, the admissible lengths include 11 and it survives the display; with the floor at 5 and $m = 1$, length 3 survives; with the floor at 18, length 11 no longer appears, since the 17-cycle's least element is below it. The cycle equation of Lemma 2 is verified in exact rationals on all three cycles, and Lemma 1's run length and identity on every odd $y < 4000$.

7. What this note does not do

It excludes no Juggler cycle. The Juggler's cycle words are, letter for letter, the words of the negative Collatz cycles (Paper A, Section 5.9, and `CollatzBridge.lean`), but a word shape does not transport a realization, so a $3n - 1$ m -cycle theorem constrains Juggler cycle words and nothing more. It does not settle the $3n - 1$ cycle question, m being bounded, and it says nothing about divergence on either map. Its fifth step, lattice reduction at $m = 50$ and beyond, is not done; Hercher's refinement is not transposed; the Eliahou-type lattice of admissible periods on this side is not written. Lemmas 1–3 are elementary and are not yet in Lean; the negative-side finance and word shape that surround them are (`neg_cycle_finance`, `neg_prefix_noncontracting`, `neg_cycle_expanding`).

8. Verification

`python -m research.juggler_sequence.negative_m_cycles` recomputes both tables in a few seconds and writes `data/research/juggler/negative_m_cycles/summary.json`; `tests/research/juggler_sequence/test_negative_m_cycles.py` checks Lemma 1 on every odd $y < 4000$, the cycle equation on the three cycles, the tightness of Lemmas 1 and 3 at 17, that the known cycles pass, that K_3 is a ceiling, that the three-gap walk agrees with a direct scan, and that the archived tables reproduce. The floor's certificate is with the branch `negative_floor_3x1`.

References

- [H23] C. Hercher, “There are no Collatz m -cycles with $m \leq 91$,” *J. Integer Seq.* 26 (2023), Article 23.3.5. arXiv:2201.00406.
- [S07] J. L. Simons, “A simple (inductive) proof for the non-existence of 2-cycles of the $3x+1$ problem,” *J. Number Theory* 123 (2007), 10–17. Section 6 treats $3x - 1$.
- [S08] J. L. Simons, “On the (non-)existence of m -cycles for generalized Syracuse sequences,” *Acta Arith.* 131 (2008), 217–254.
- [SdW] J. L. Simons and B. M. M. de Weger, “Theoretical and computational bounds for m -cycles of the $3n+1$ -problem,” *Acta Arith.* 117 (2005), 51–70.
- [Si05] J. L. Simons, “On the nonexistence of 2-cycles for the $3x+1$ problem,” *Math. Comp.* 74 (2005), 1565–1572.
- [R87] G. Rhin, “Approximants de Padé et mesures effectives d’irrationalité,” *Séminaire de Théorie des Nombres, Paris 1985–86*, Progress in Mathematics 71, Birkhäuser, 1987, 155–164.
- [LMN] M. Laurent, M. Mignotte and Y. Nesterenko, “Formes linéaires en deux logarithmes et déterminants d’interpolation,” *J. Number Theory* 55 (1995), 285–321.
- [Si03] M. K. Sinisalo, “On the minimal cycle lengths of the Collatz sequences,” preprint, University of Oulu, 2003.
- [E93] S. Eliahou, “The $3x+1$ problem: new lower bounds on nontrivial cycle lengths,” *Discrete Math.* 118 (1993), 45–56.
- [St77] R. P. Steiner, “A theorem on the Syracuse problem,” *Proc. 7th Manitoba Conf. Numerical Math.* (1977), 553–559.

- [L85] J. C. Lagarias, “The $3x+1$ problem and its generalizations,” *Amer. Math. Monthly* 92 (1985), 3–23; and the annotated bibliographies, arXiv:math/0309224 and math/0608208.
- [A] P. Cochin, “Lower Bounds for Cycle Lengths in the Juggler Map” (Paper A), Section 5.9, and `formal/Problems/Juggler/CollatzBridge.lean`, in the repository <https://github.com/sneakyweasel/btlab/>.
- OEIS A037084, comment on the $3x - 1$ verification to 10^8 .